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Valladolid

Machine Learning tools to assist the analysis of complex rotational spectra of chemical mixtures

D. Yélamos, Z. Wang, C. Calabrese, A. Lesarri, C. Bermúdez

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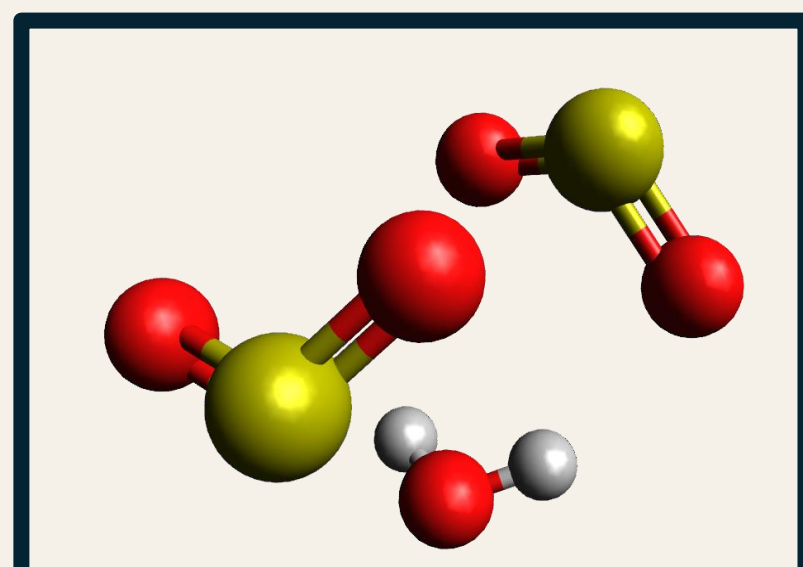
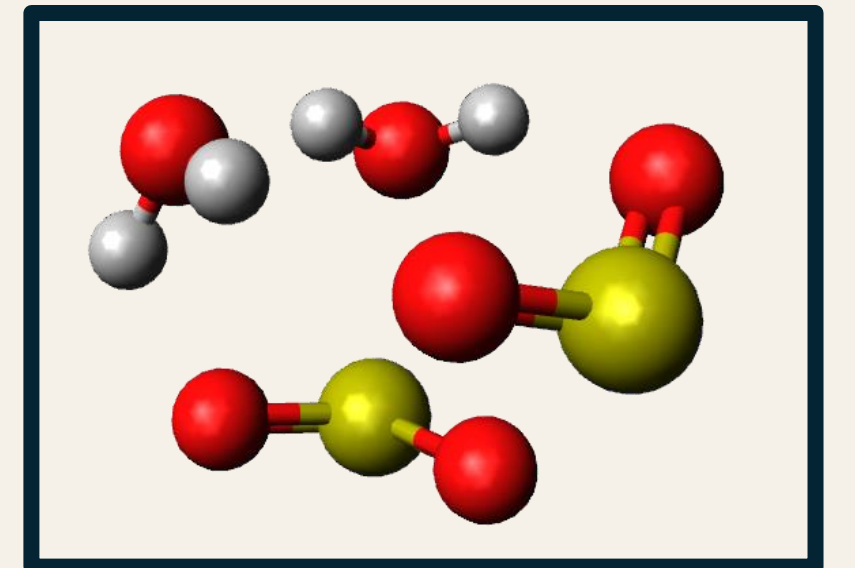
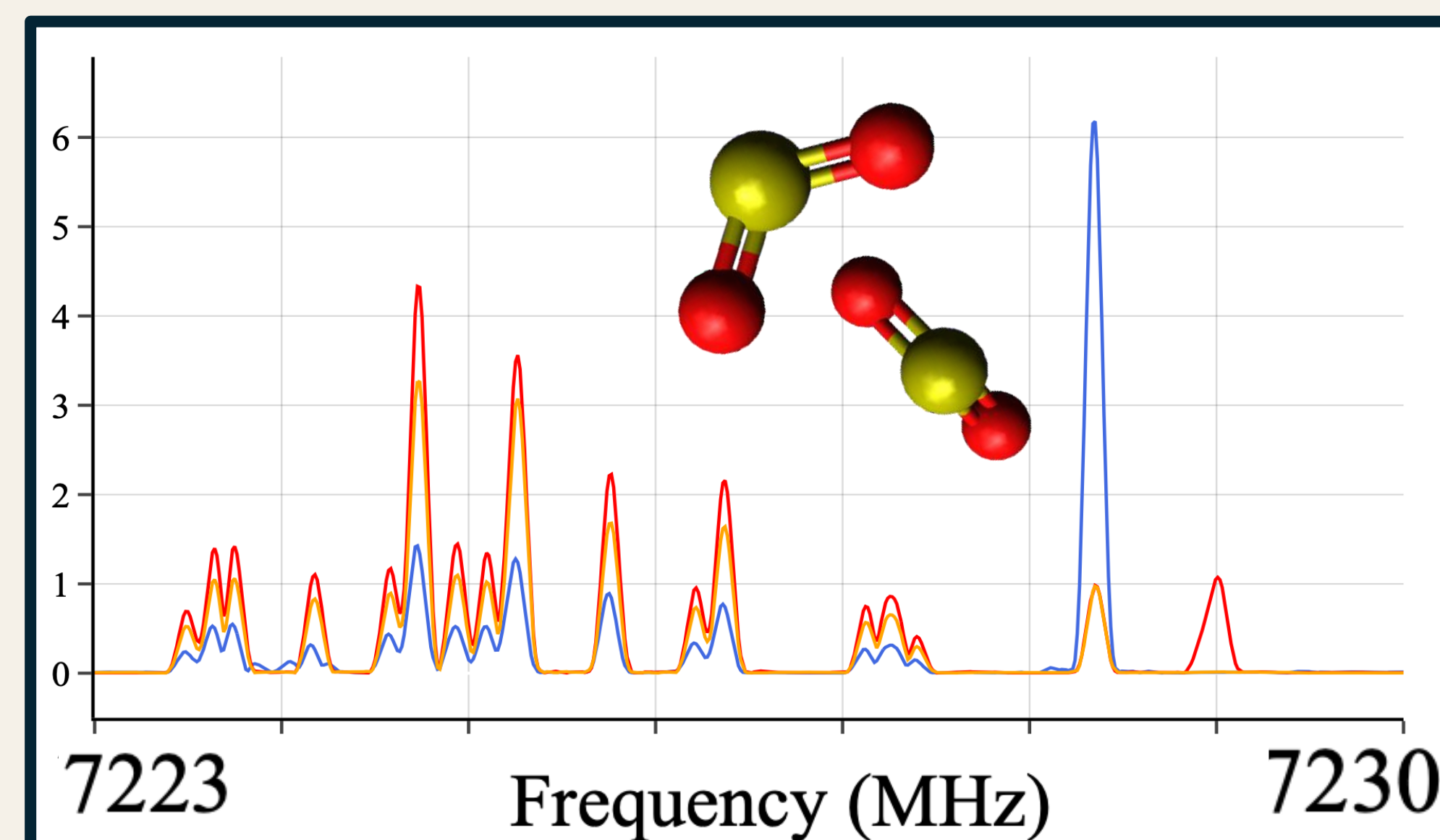
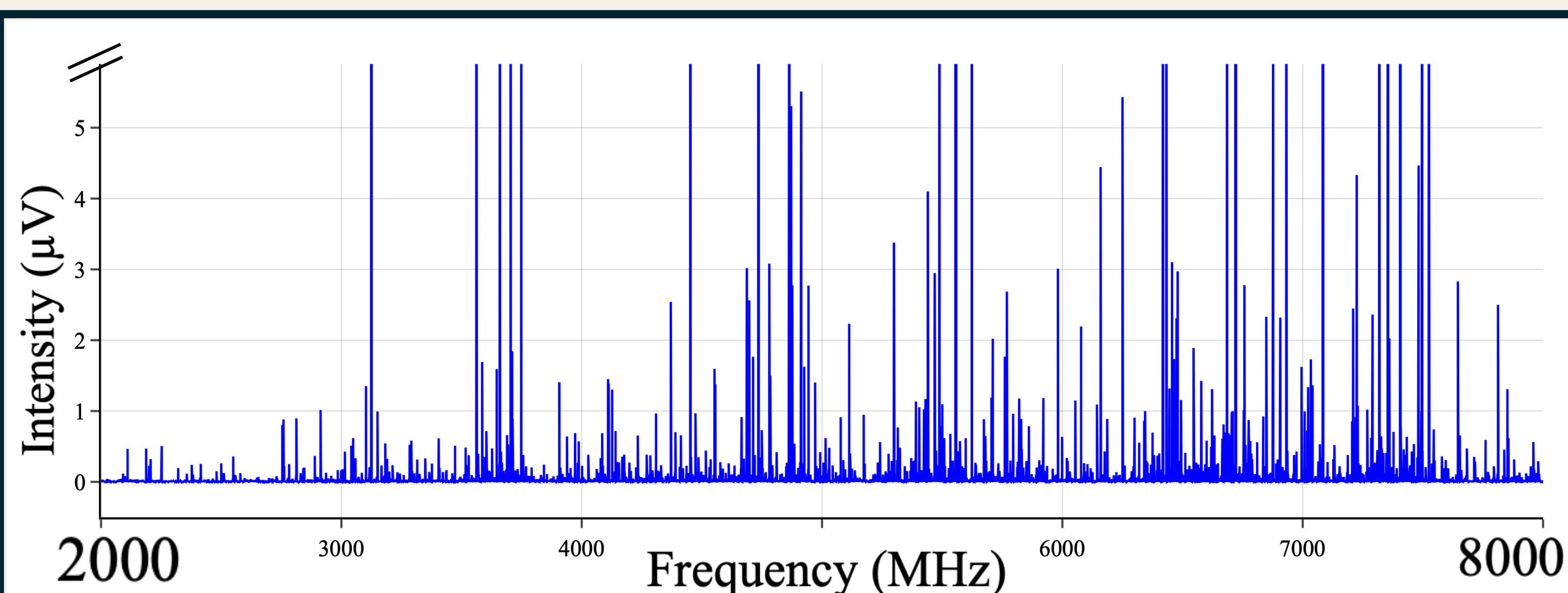
Introduction The possibility of averaging millions of spectra thanks to Broadband chirped pulse Fourier Transform microwave spectroscopy, makes rotational analysis challenging involving thousands of transitions lines. New methods have arisen to facilitate the analysis of the molecular spectra, such as frequency-based methods [1]. Within this study we focus on intensity-based methods for chemical mixtures ($\text{SO}_2/\text{H}_2\text{O}/\text{D}_2\text{O}$), where we use the unique response of each chemical species to change in composition. We explore new approaches using Machine Learning tools to disentangle rotational spectra.

Experimental We measured rotational spectra of three different mixtures of gases using Broadband CP-FTMW spectrometer placed in Universidad de Valladolid.

SO_2 pure

+ H_2O

+ D_2O



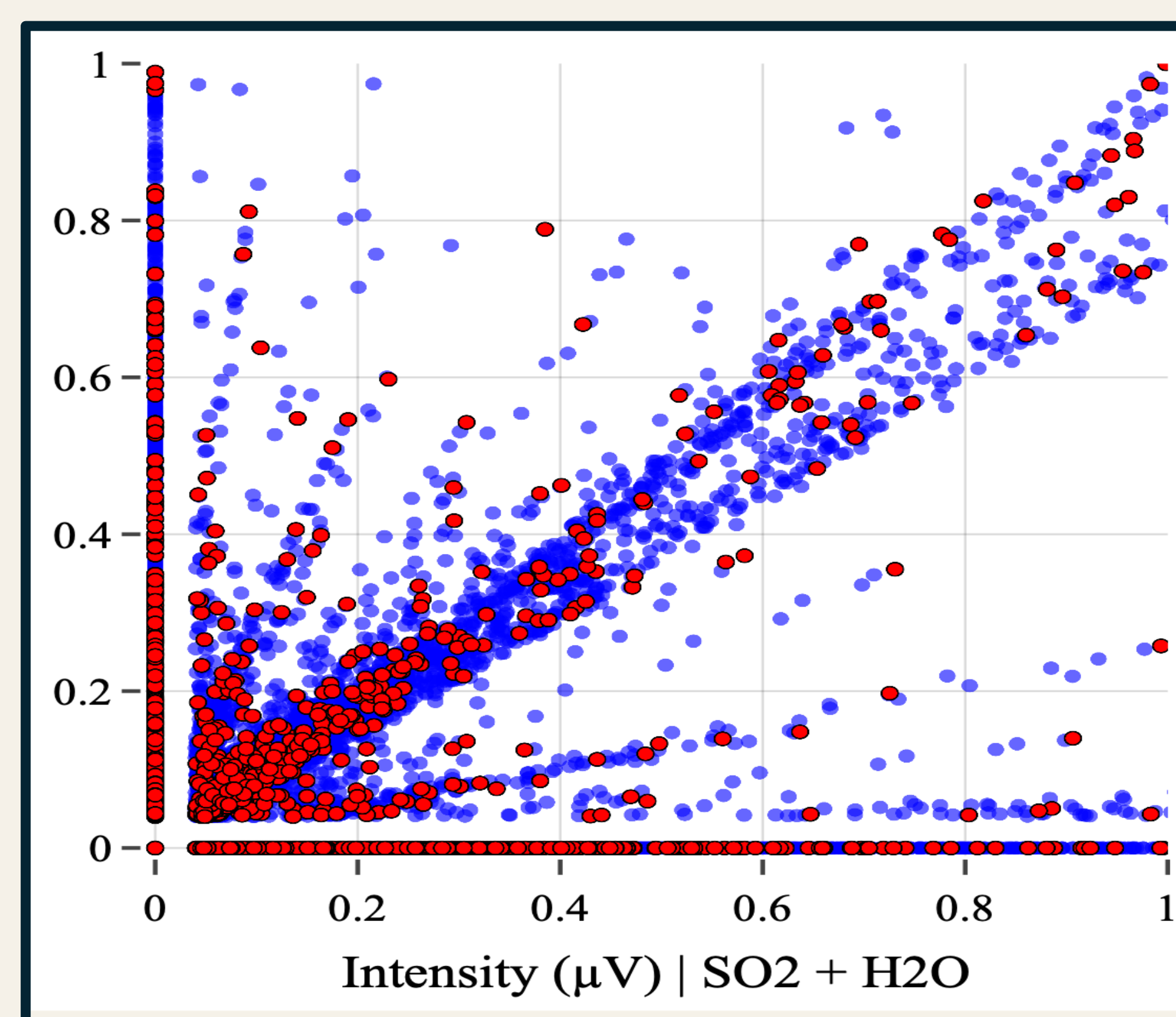
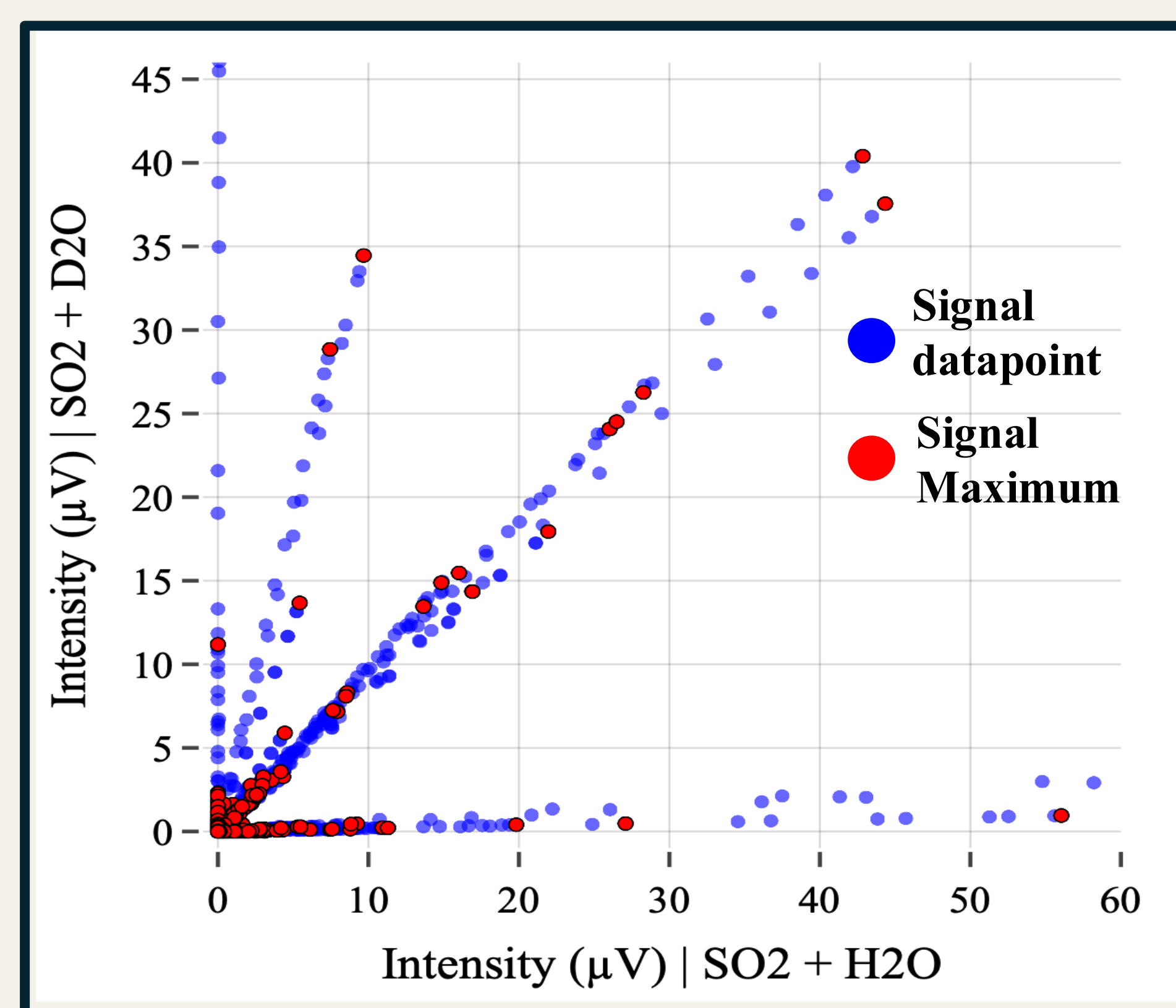
Methods By facing intensities of the spectra of two mixtures with different composition [2], we detect linear patterns called “rays”. Each ray corresponds to a unique response to changes in composition and therefore to a specific chemical specie. We developed different unsupervised clustering Machine Learning algorithms to isolate each ray and assign each cluster to a chemical species of $\text{SO}_2 / \text{H}_2\text{O}$. We focus on the best performance achieved by RANSAC (Random sample consensus) algorithm.

Modified RANSAC Algorithm

1. Iterates over all possible linear regressions forcing the intercept to origin.
2. Scores regressions according to the number of inliers within an angular threshold.
3. Clusters all inliers within the best regression.
4. Removes this cluster and repeats the process.

Hyperparameters

- Minimum datapoints used per regression = 2
- Maximum number of clusters = 20
- Maximum iterations = 10 000
- Angular threshold = 0.02 rads

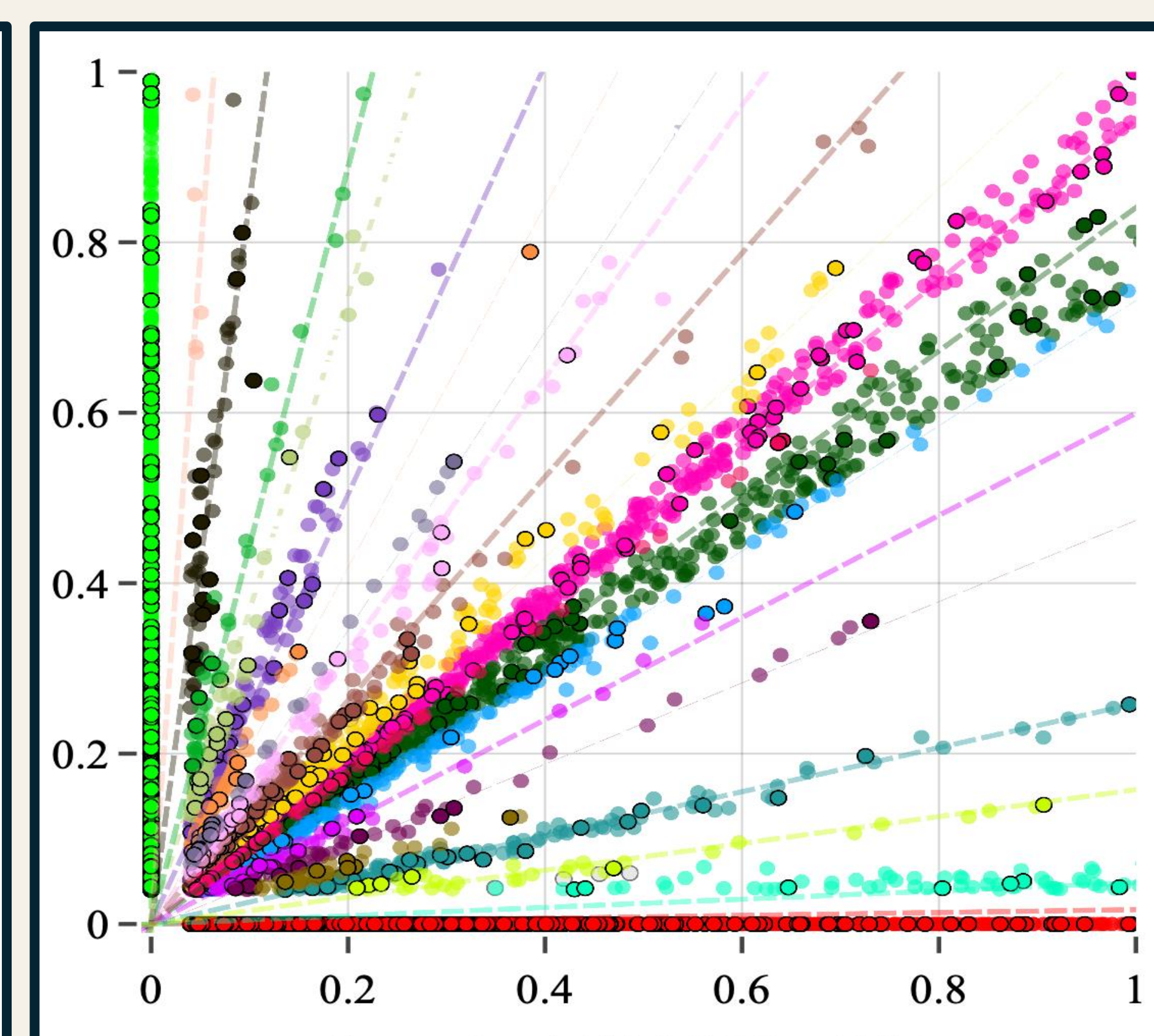
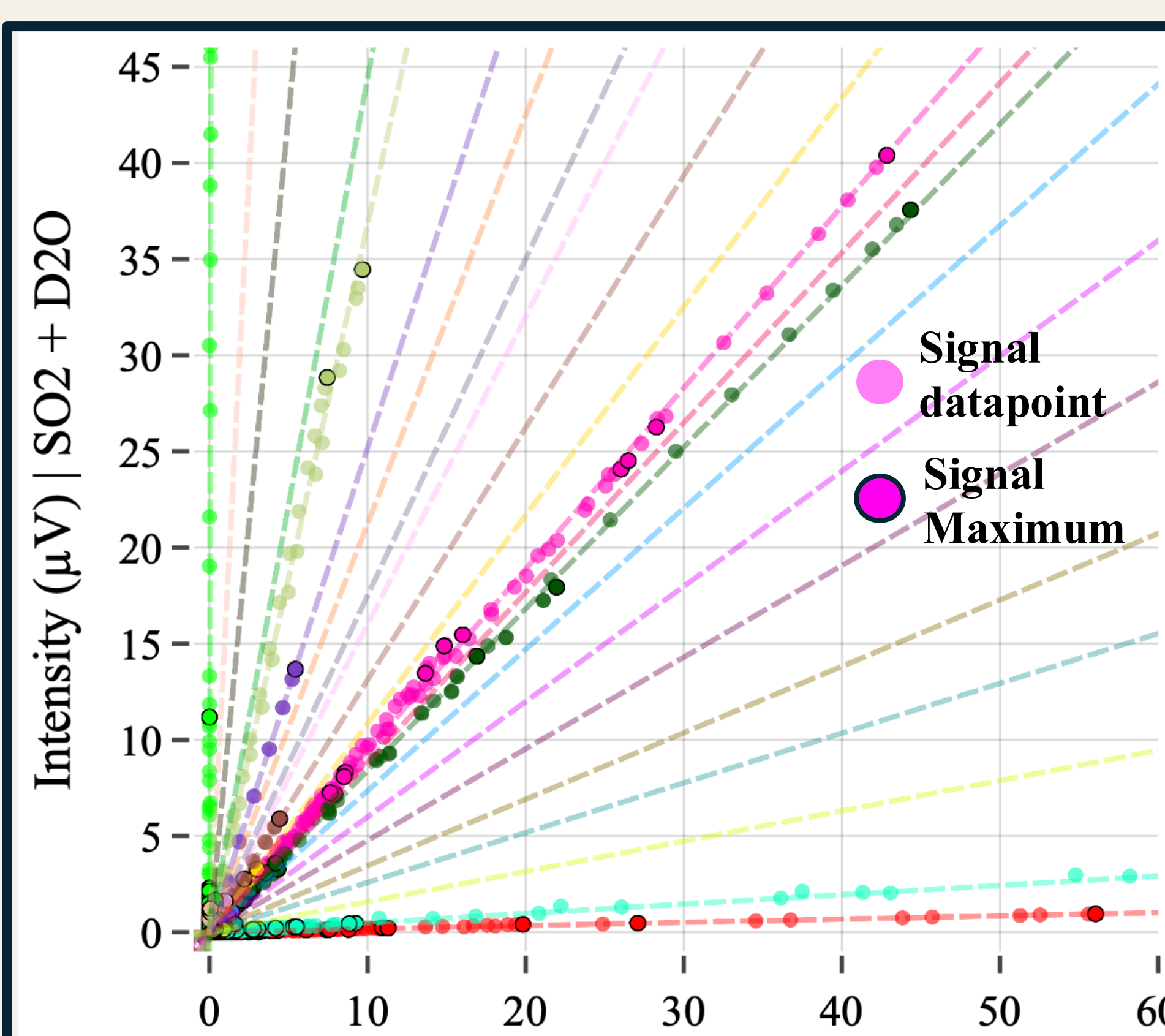


Discussion After trying other model's algorithm like DBSCAN, FLEMIX and GMM we consider RANSAC algorithm to be best suited due to its hyperparameter flexibility. RANSAC shows great results. However it is sensitive to high densities of data and distance threshold. Hyperparameters adjustment is always required for specific problems.

How to deal with these drawbacks:

- Work only with maximums as data to avoid high densities overfitting.
- Analyze distributions of each datapoint distance to each ray slope.

Results of RANSAC model

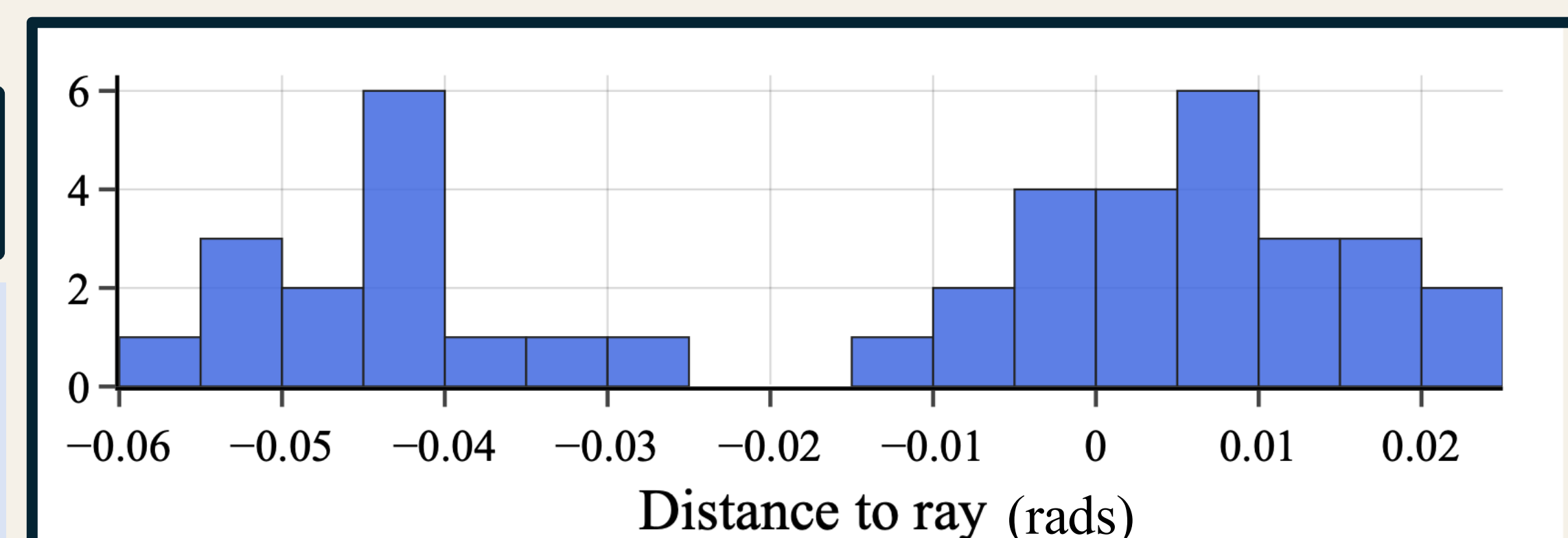
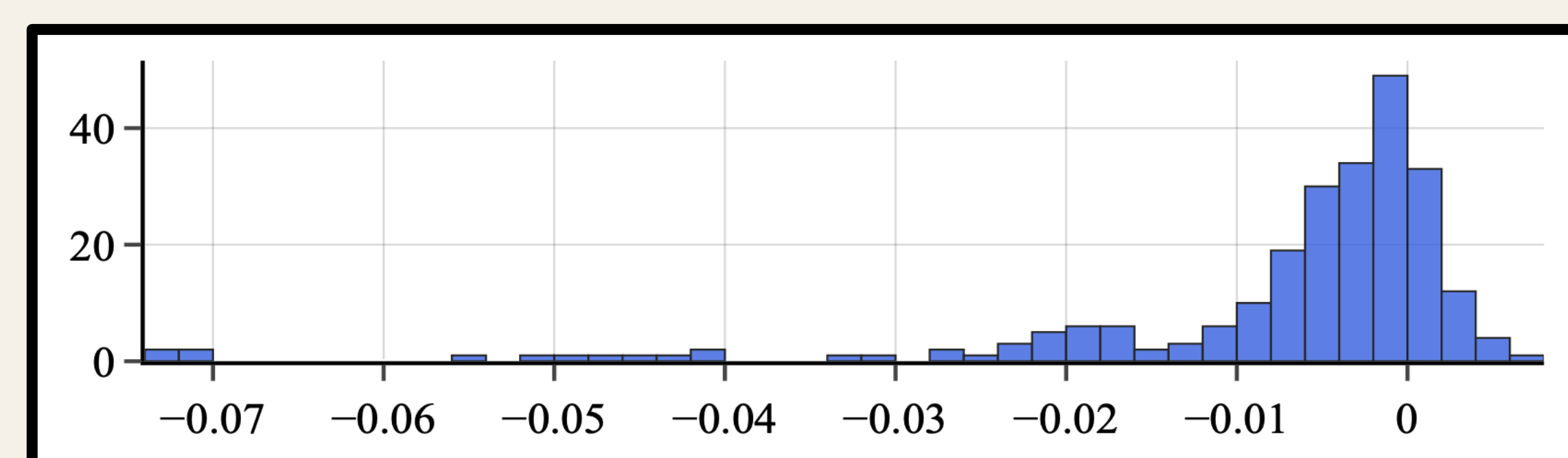


References

- [1] N. A. Seifert et al., Mol. Spectrosc. 321, 13-21 (2015).
[2] K. C. Gilbert et al., J. Phys. Chem. A. 129 (45), 10393-10403 (2025).

Acknowledgments

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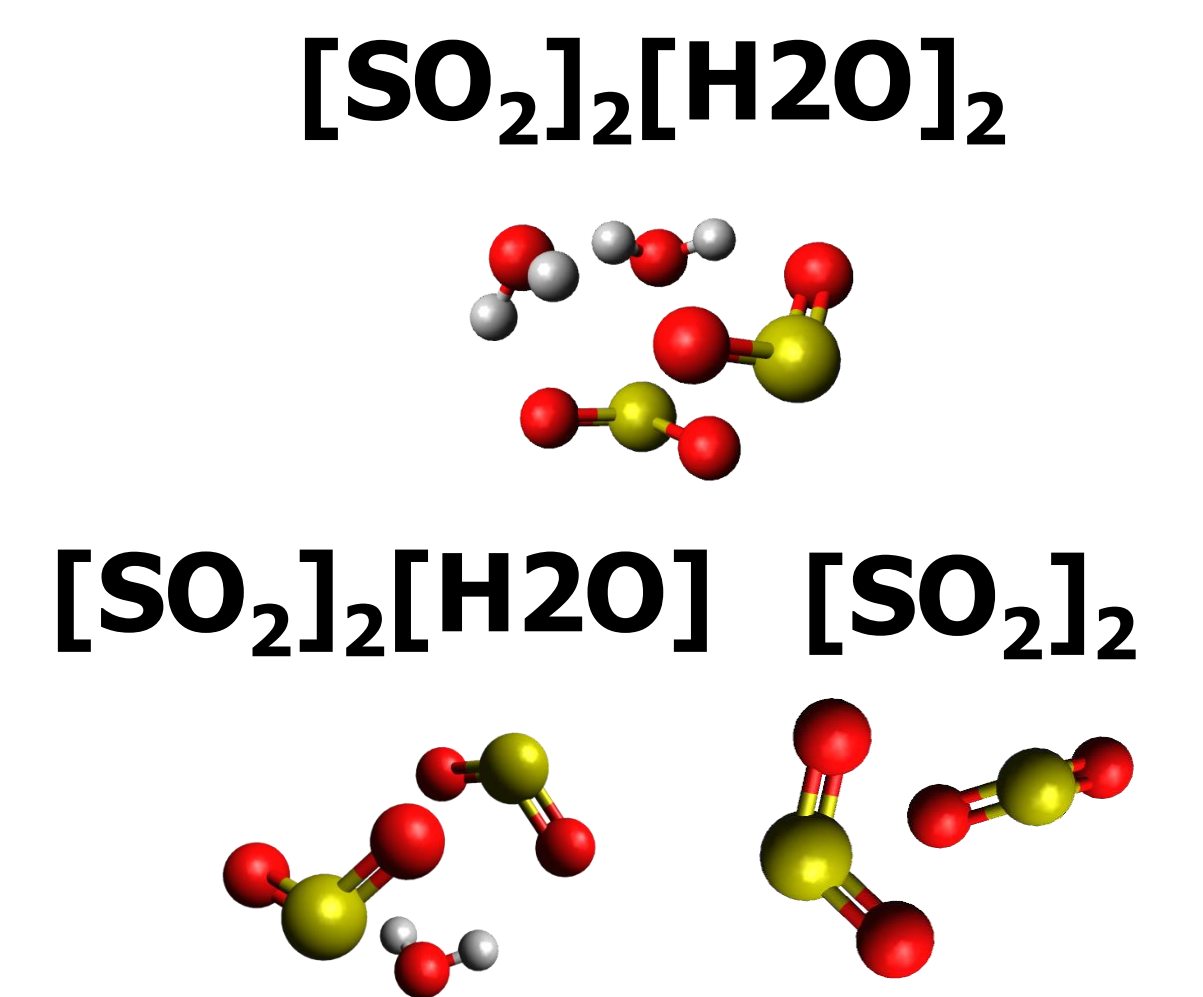
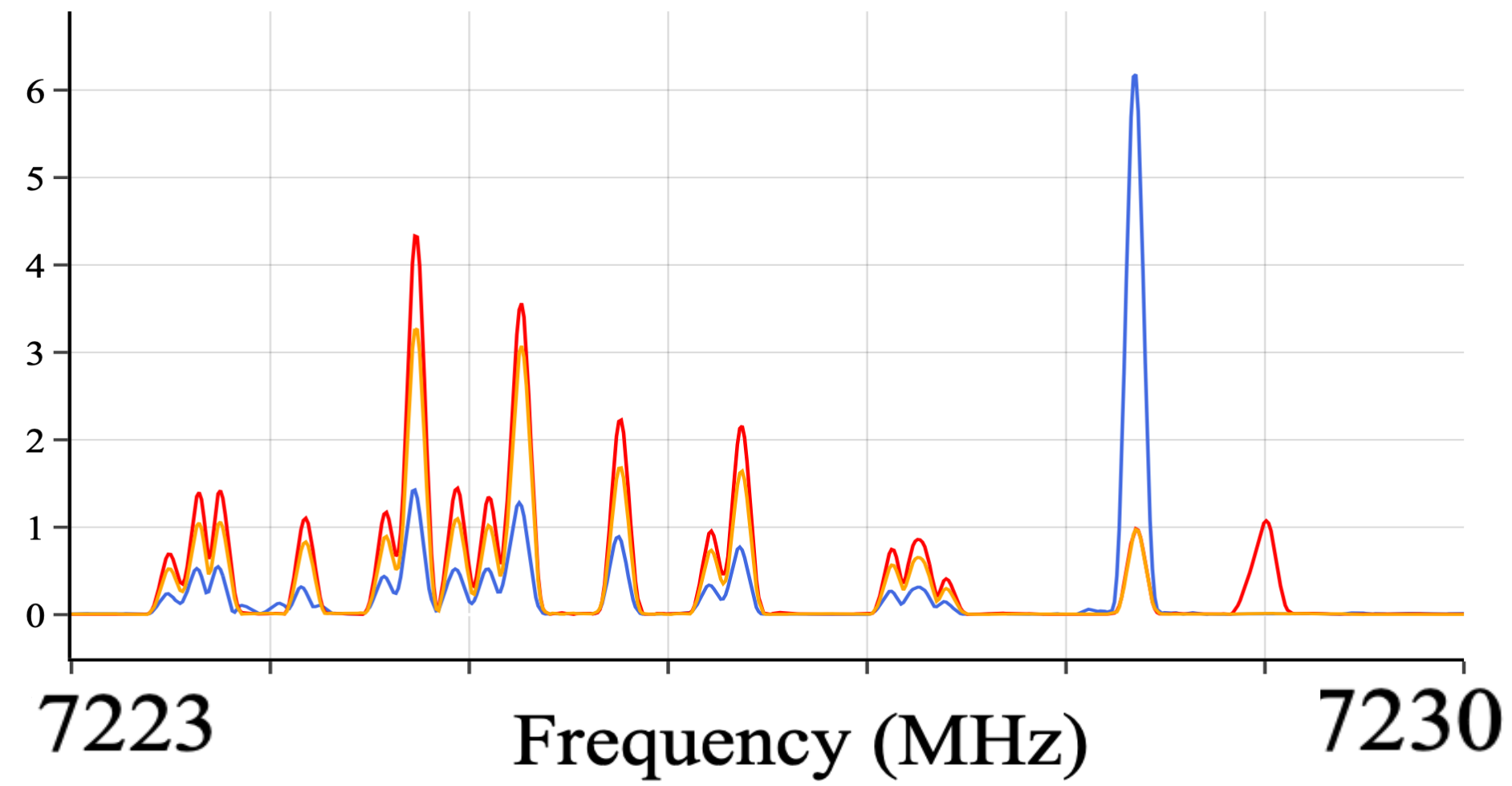
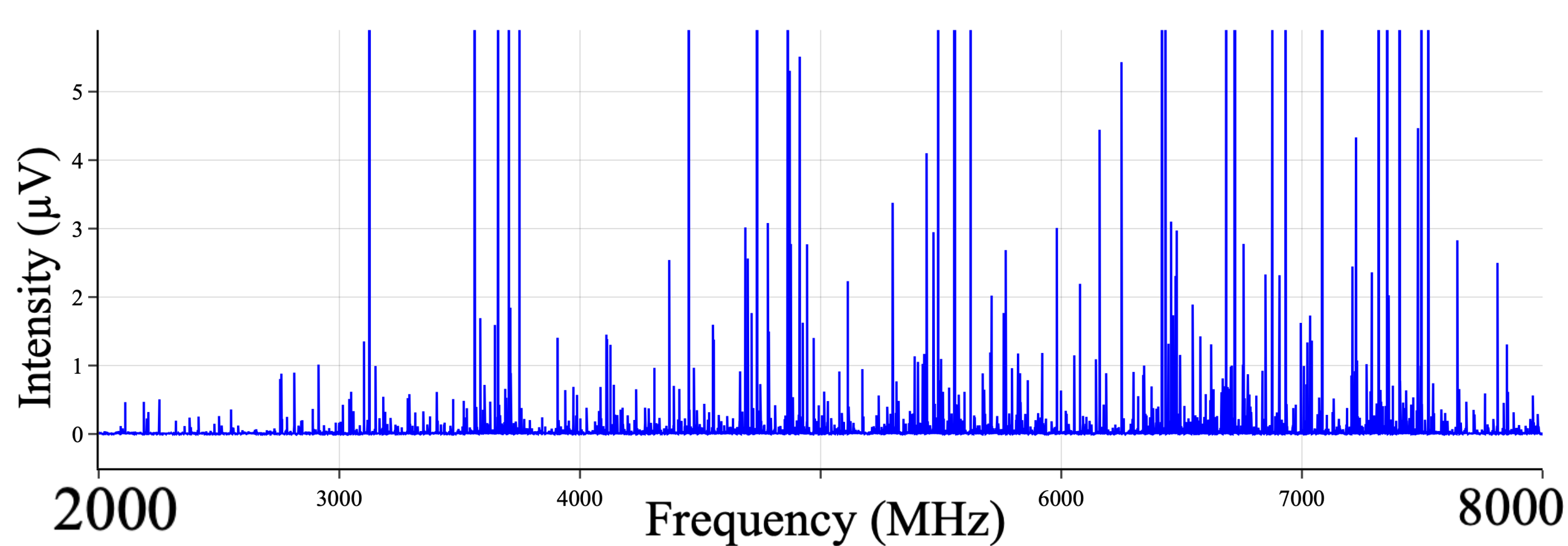
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Experimental We measured rotational spectra of three different mixtures of gases using Broadband CP-FTMW spectrometer placed in Universidad de Valladolid: pure SO_2 , mixture with H_2O and D_2O .

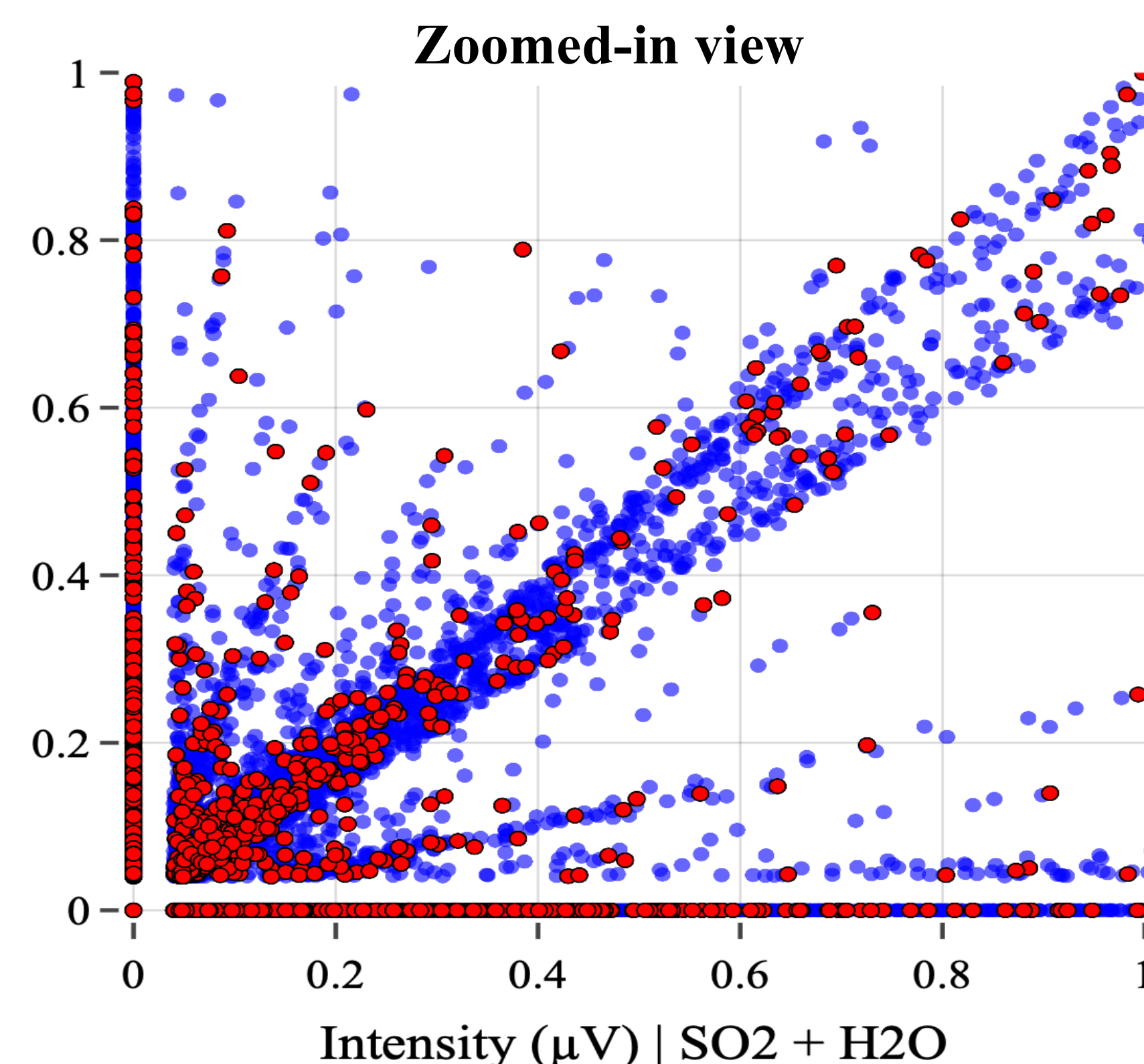
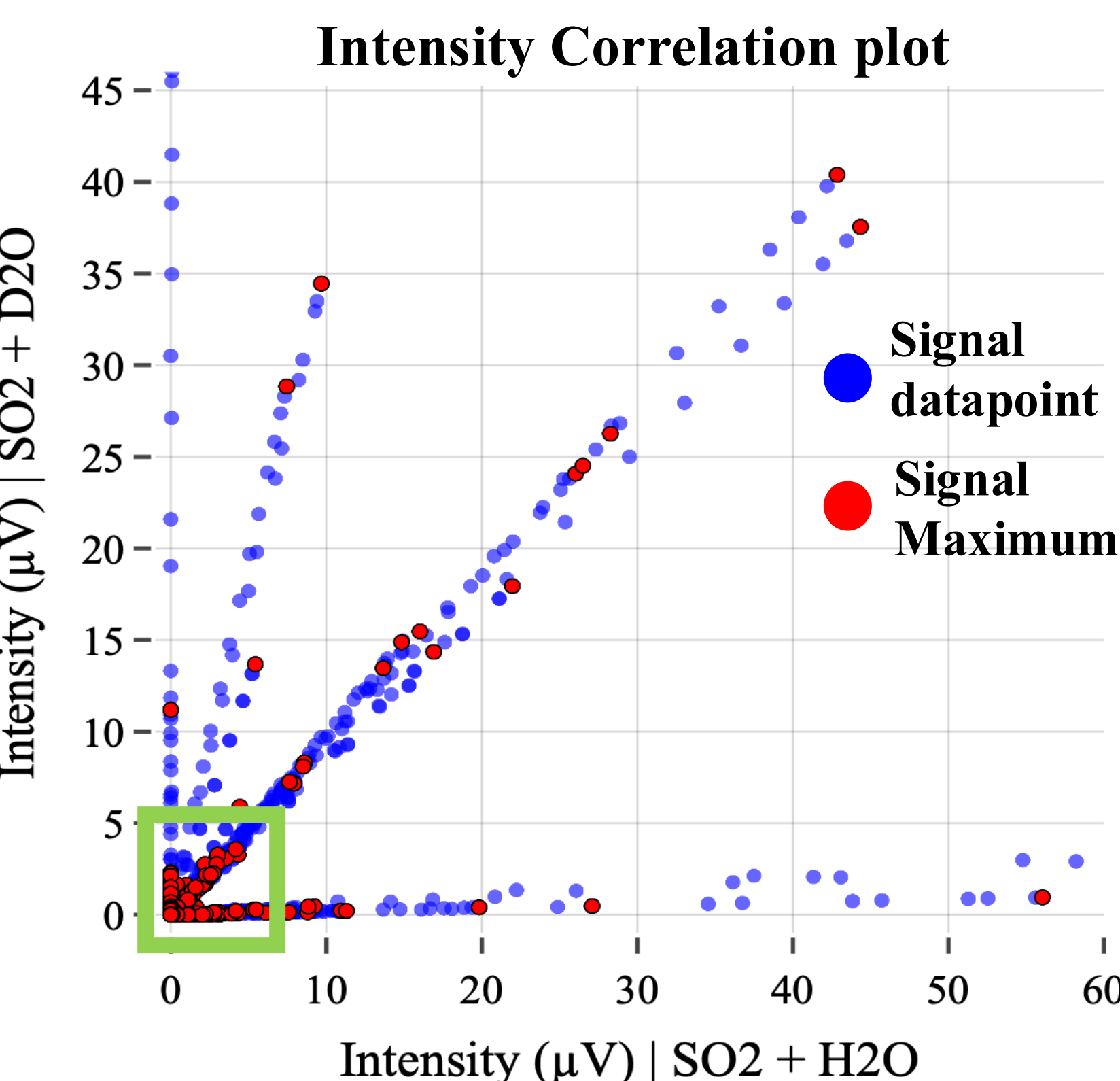
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Methods By facing intensities of the spectra of two mixtures with different composition [K. C. Gilbert et al., J. Phys. Chem. A. 129 (45), 10393-10403 (2025)], we detect linear patterns called “rays”. Each ray corresponds to a unique response to changes in composition and therefore to a specific chemical specie. We developed different unsupervised clustering Machine Learning algorithms to isolate each ray and assign each cluster to a chemical species of $\text{SO}_2/\text{H}_2\text{O}$. We focus on the best performance achieved by RANSAC (RANDOM Sample Consensus) algorithm.



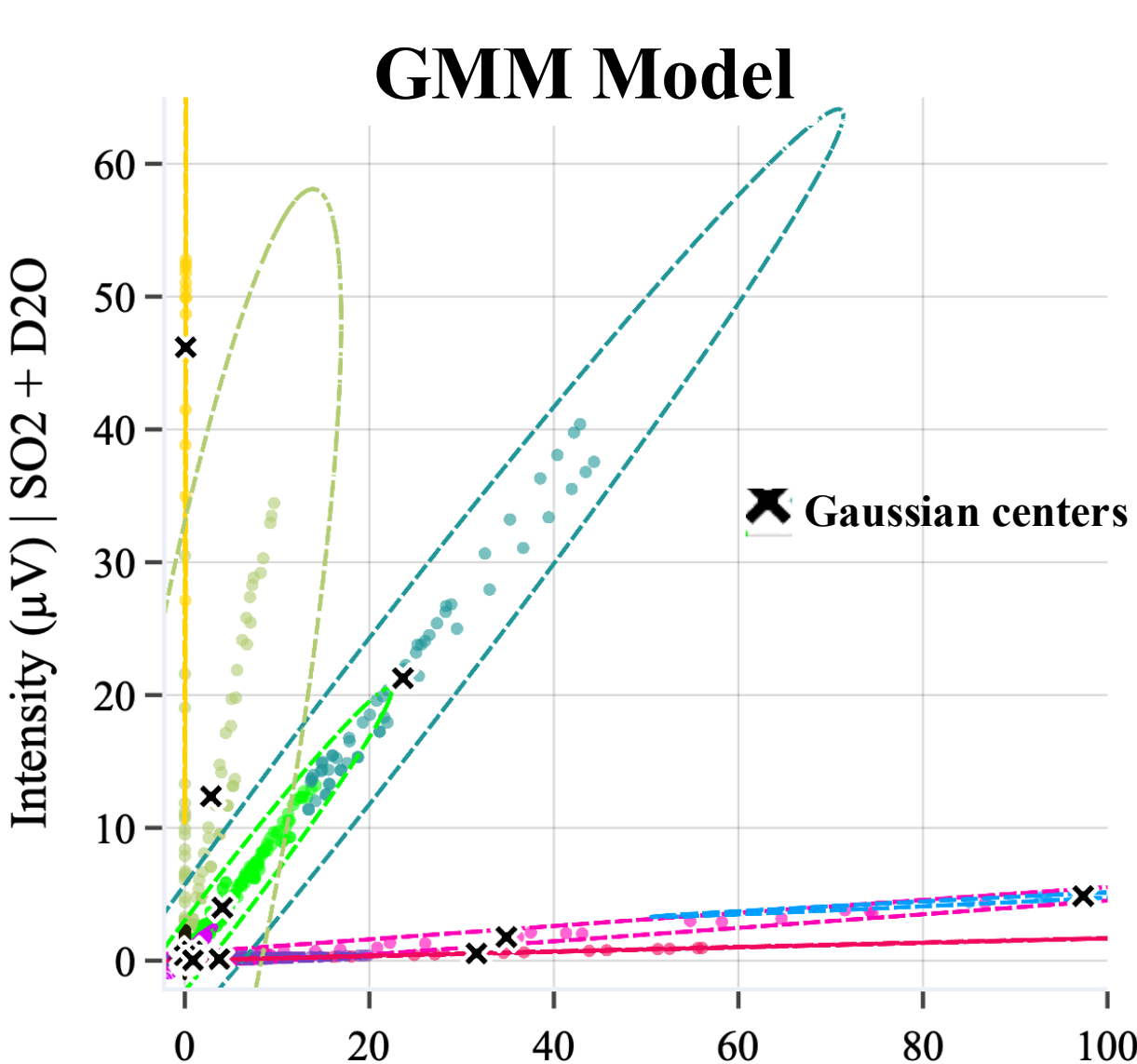
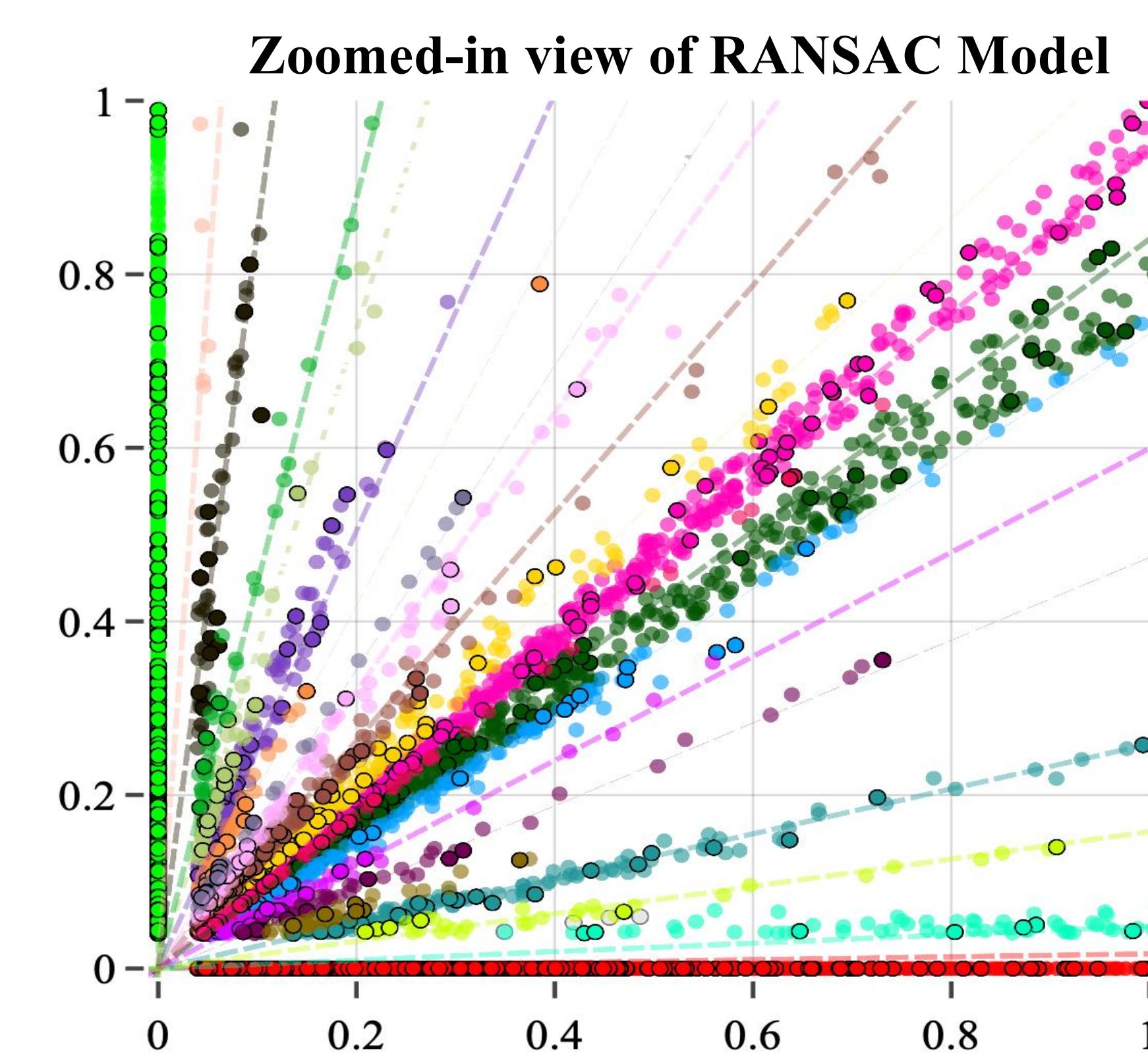
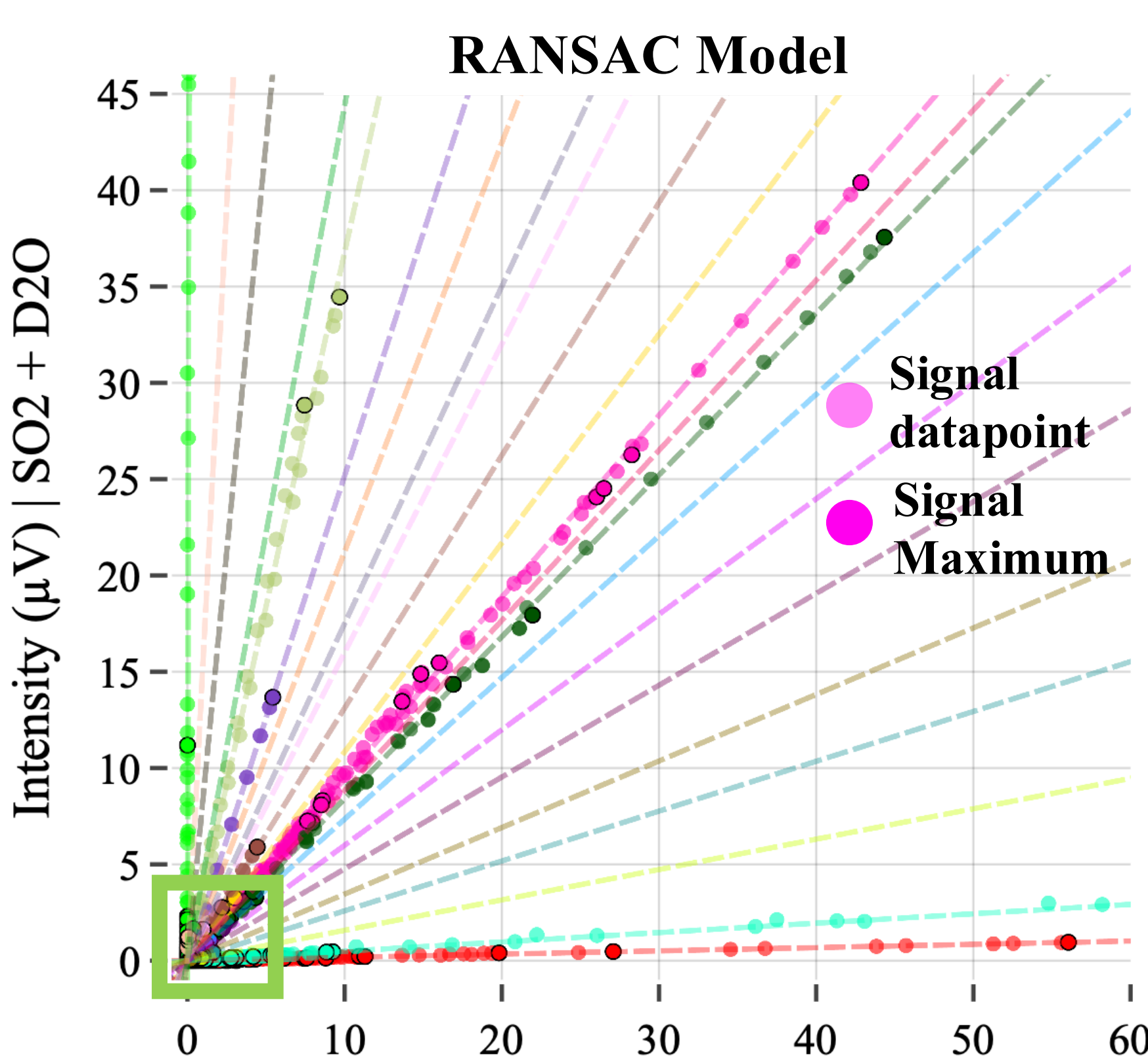
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Results - Models



Between all four models tested we present the two of our interest

- DBSCAN (Density-Based Spatial Clustering of Applications with Noise).
- FlexMix (Flexible Mixture Modeling).
- GMM (Gaussian Mixture Models).
- RANSAC

While RANSAC offers the highest flexibility in terms of hyperparameters and returns visible, a priori, good results based on ratios tendency. GMM model fails completely since it clusters based on intensity values and not slopes.

Distance distribution to rays could give hints of good clustering

Discussion

The nature of the RANSAC model makes it very sensible to high densities provoking a bias towards intense signals. It does not look for a cumulus of signals with the same slope, it tries to cluster as much points as possible, ignoring low intensity signals.

- An overfitted RANSAC clustering and a posterior clustering by slope proximity could be interesting.
- Working only with signals maximums removes high densities from intense signals.

Future prospects

1. By converting data into ratio or angular space, GMM model would not deal with absolute magnitudes
2. Increasing dimensionality of the problem:
 - Different mixture compositions.
 - Temporal evolution of each signal.

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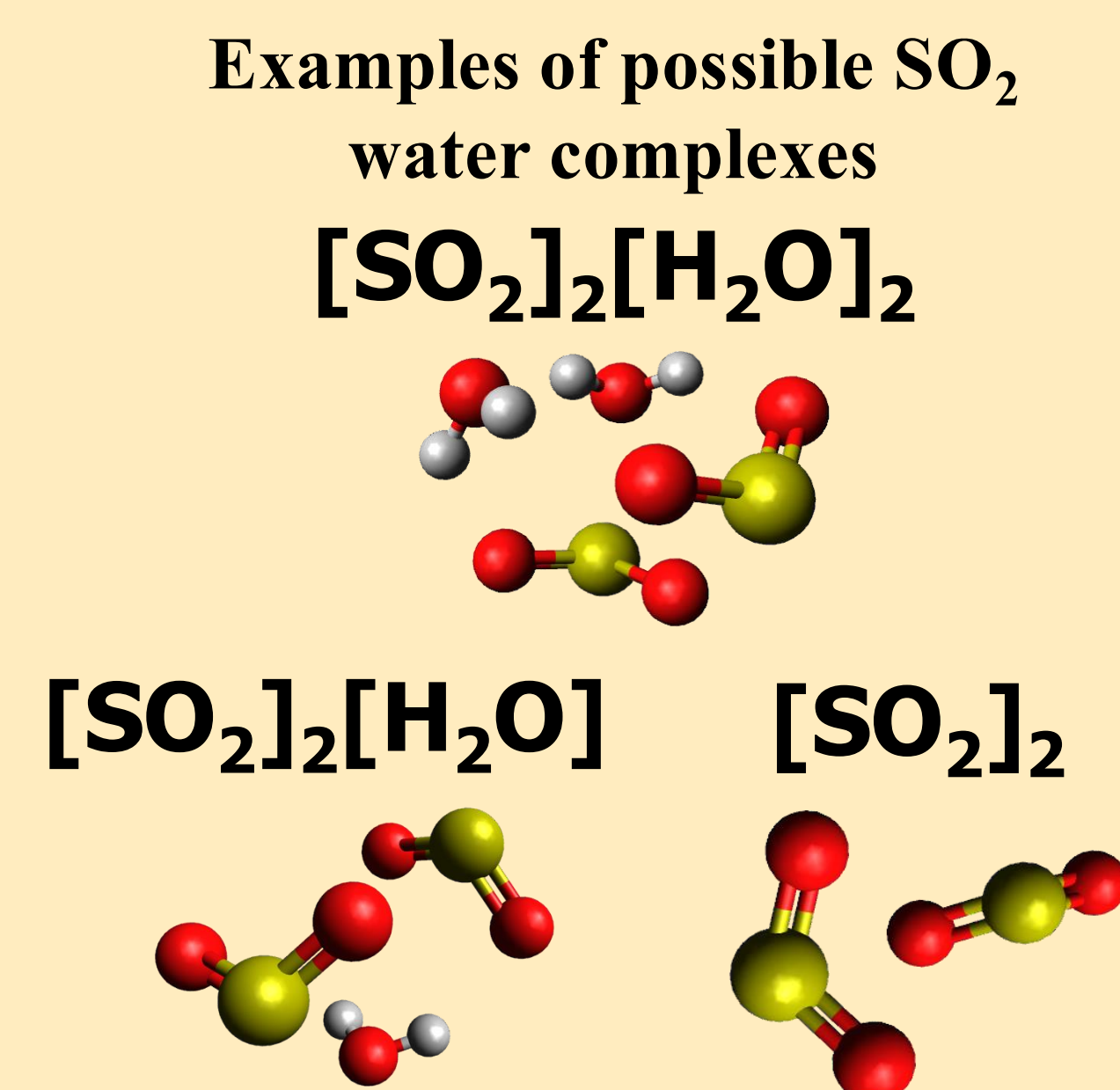
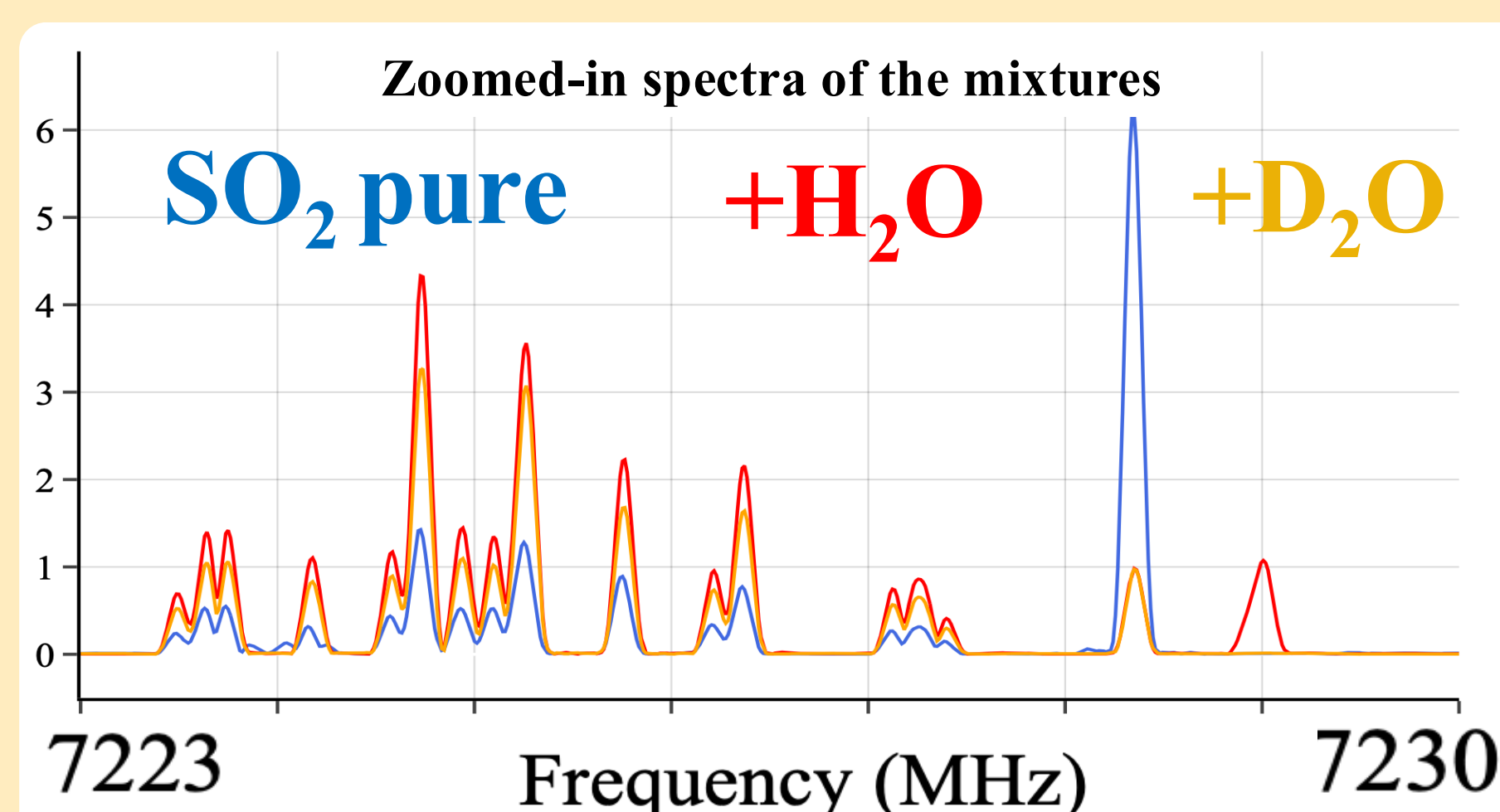
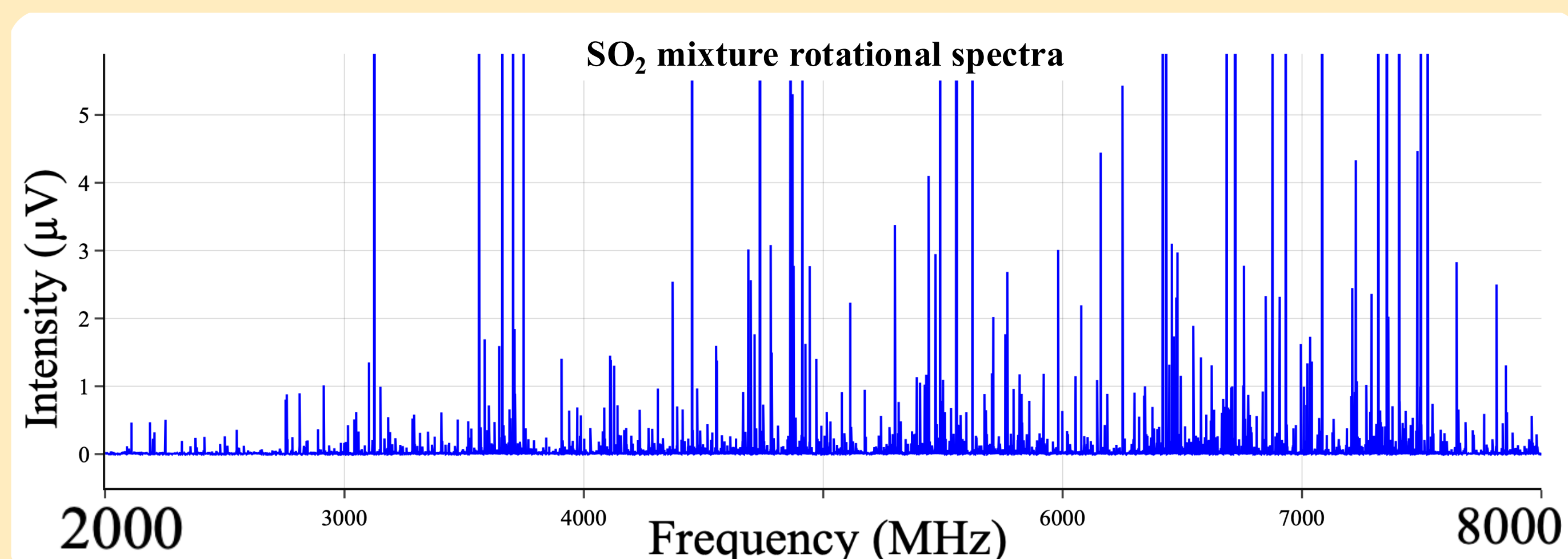
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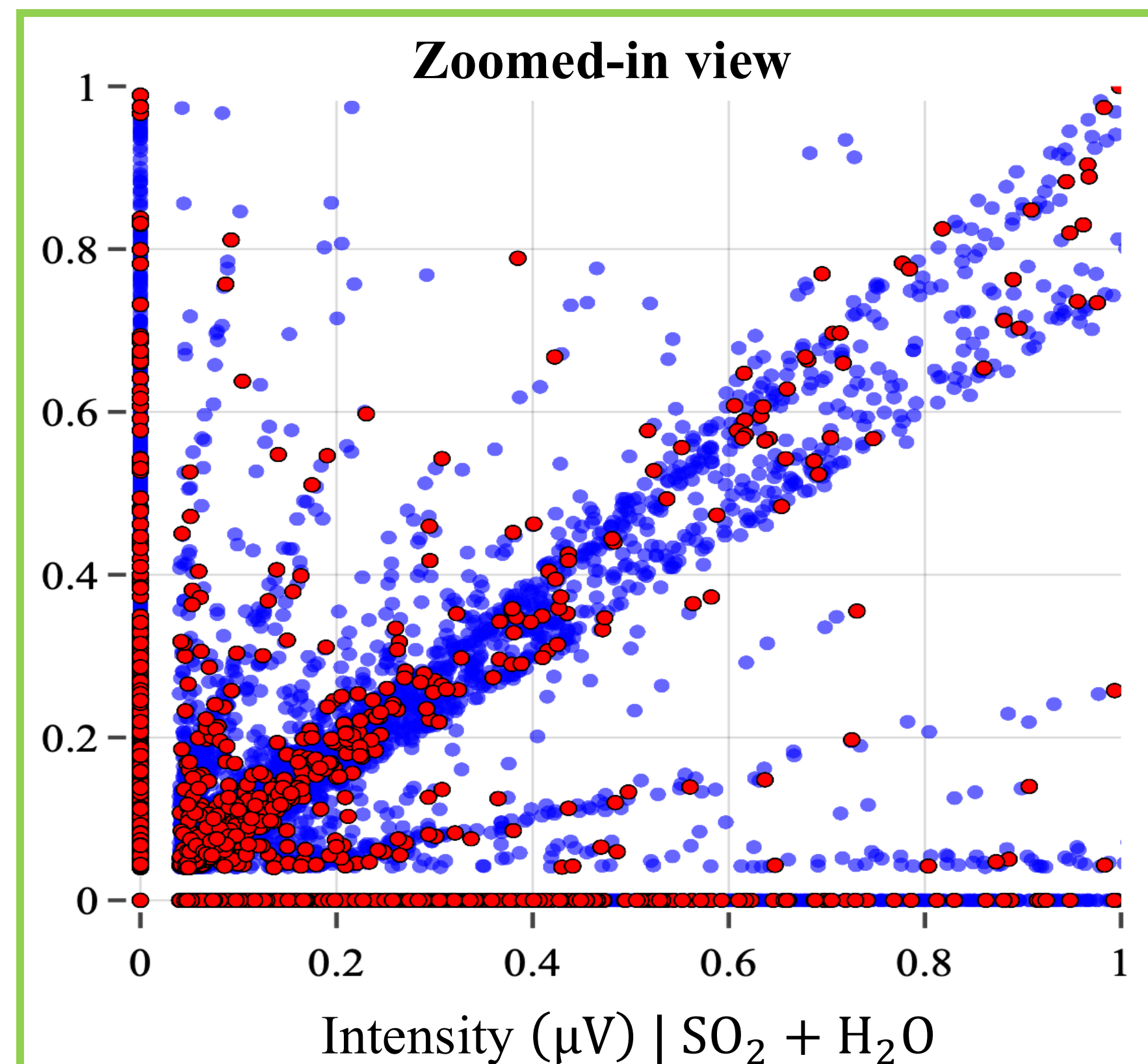
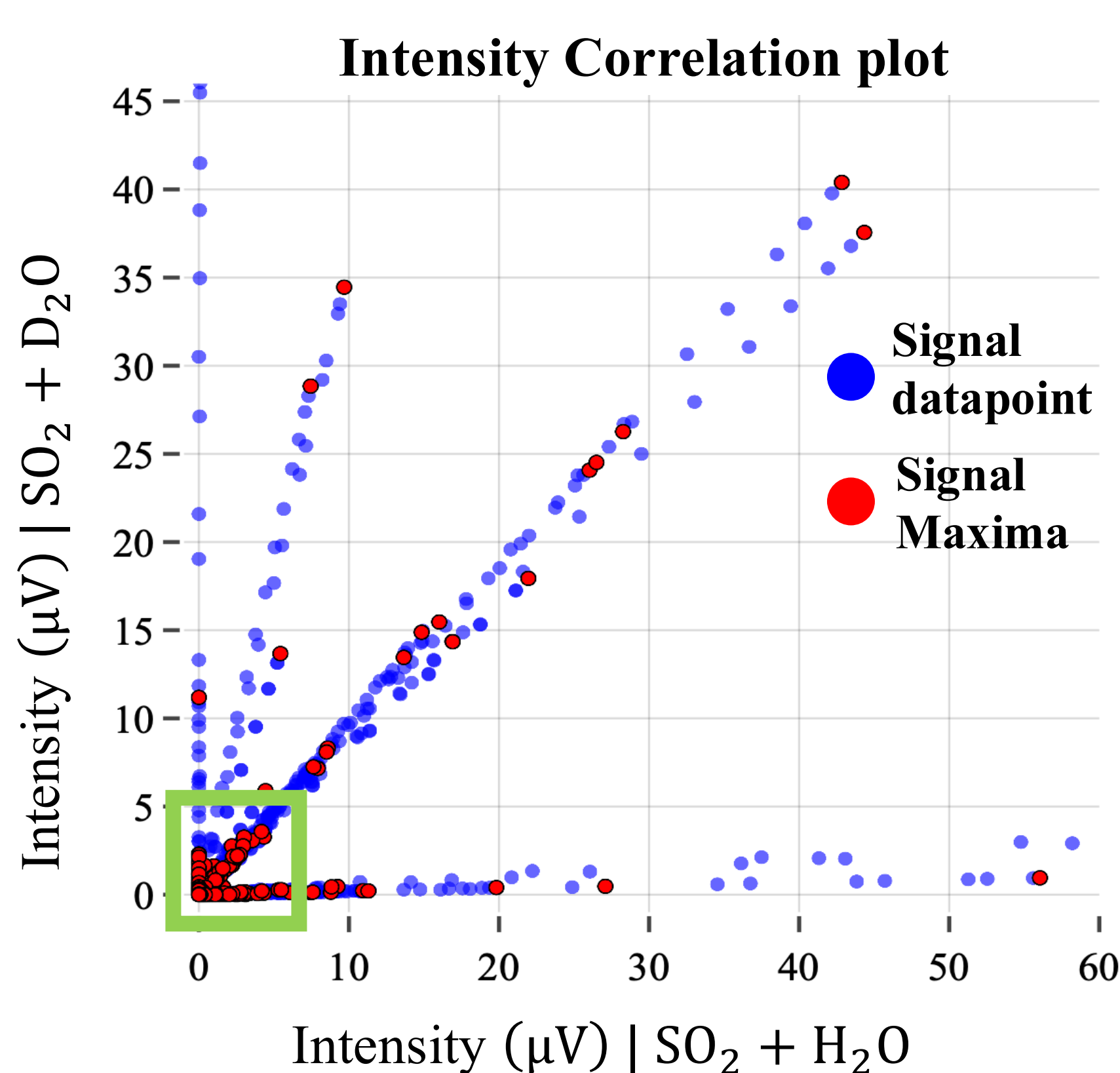
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Experimental We measured rotational spectra of three different mixtures of gases using the **broadband CP-FTMW** spectrometer placed at Universidad de Valladolid: SO₂(1% in Ne), SO₂:H₂O and SO₂:D₂O.



Methods By facing intensities of the spectra of two mixtures with different composition [Jacobson *et al.*, JChemPhys 107, 8349, (1997)], we detect linear patterns called “rays”. Each ray corresponds to a unique response to changes in composition and therefore to a specific chemical species. We developed different **unsupervised clustering Machine Learning algorithms** to isolate each ray and assign each cluster to a chemical species of SO₂/H₂O. We focus on the best performance achieved by **RANSAC** (Random Sample Consensus) algorithm.



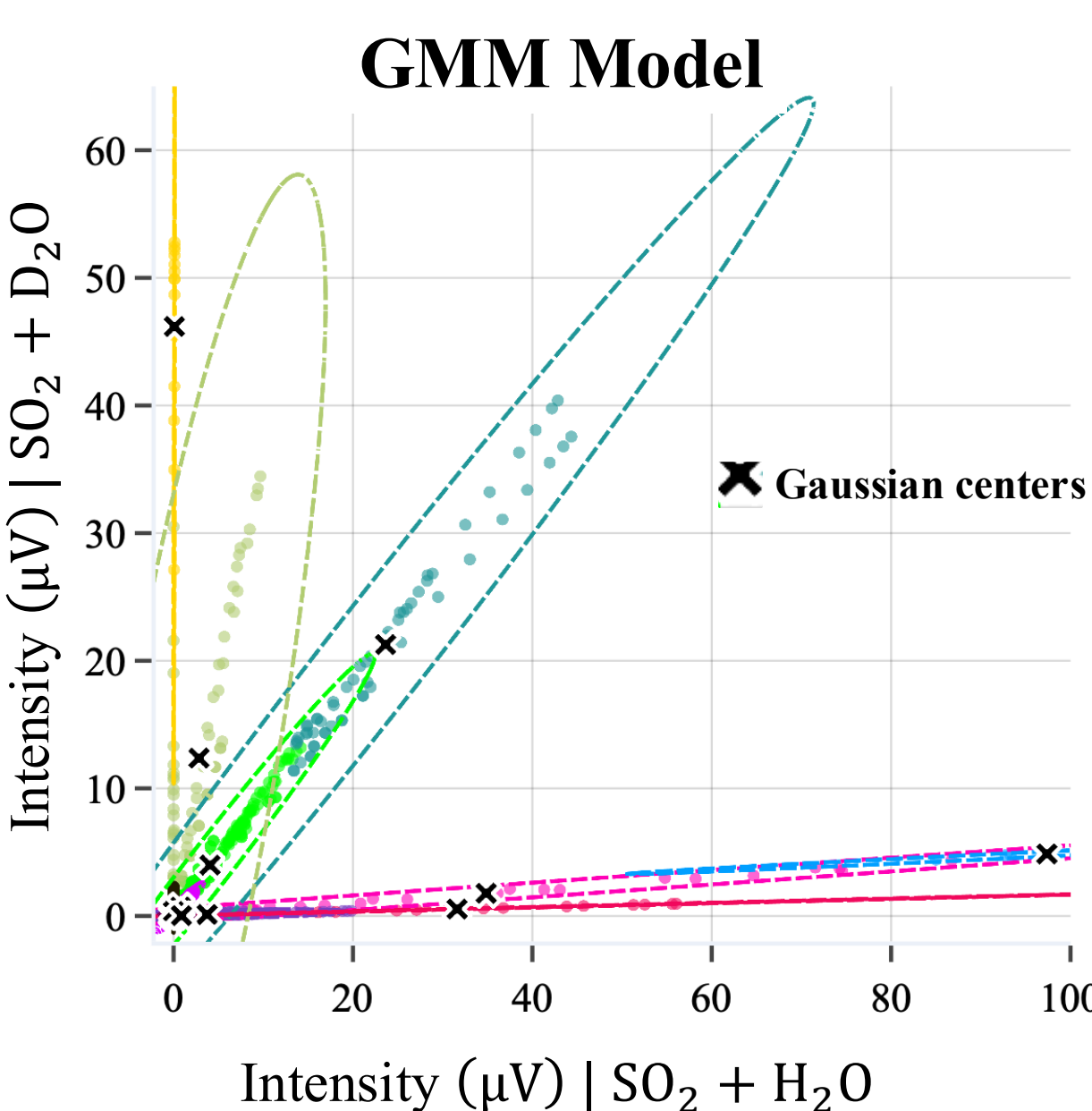
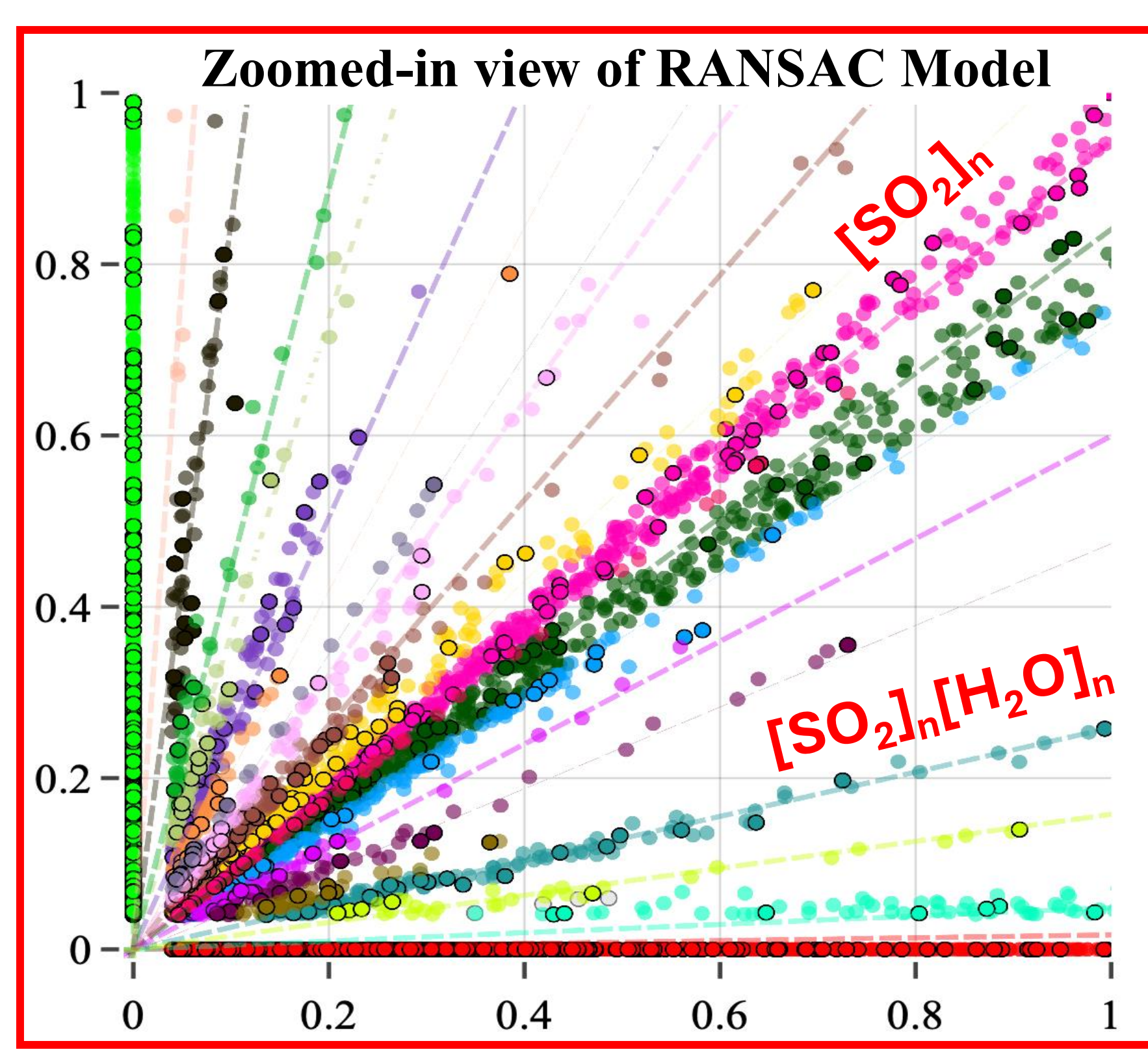
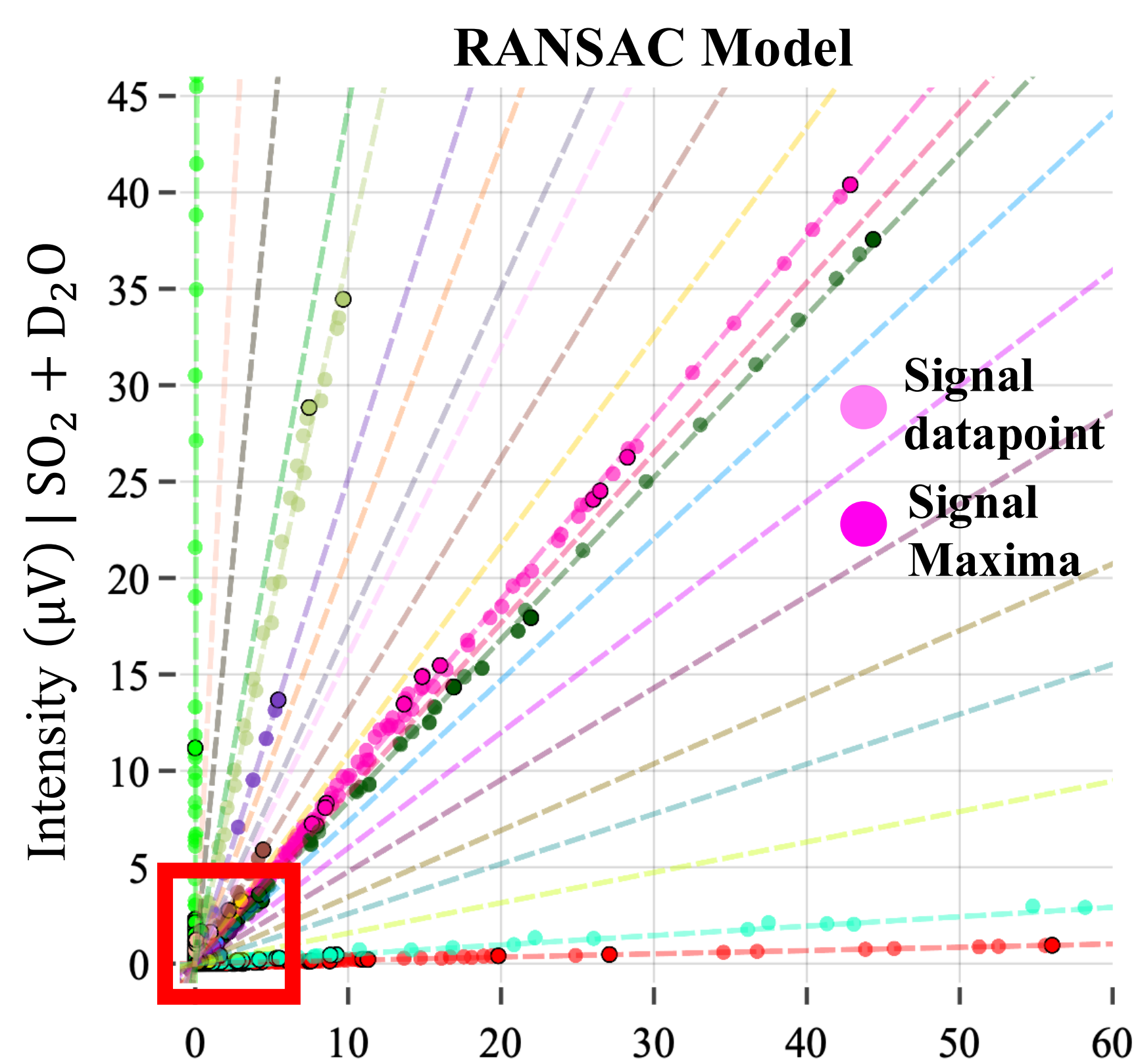
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- **RANSAC.**

While RANSAC offers the **highest flexibility** in terms of hyperparameters and returns visible, a priori, good results based on tendency ratios. GMM model fails completely since it clusters based on intensity values and not slopes.

Discussion

RANSAC model is able to distinguish clusters that corresponds to the different SO₂ complexes. Each ray correspond to the complexes with the same behavior in presence of water, hence the same amount of water in their structure. The nature of the model makes it **very sensible to high densities leading to a bias towards intense signals**. It does not look for a cumulus of signals with the same slope, it tries to cluster as much points as possible, ignoring low intensity signals.

- An overfitted RANSAC clustering and a **posterior clustering** by slope proximity could be interesting.
- Working **only with signals maxima** removes high densities coming from intense signals.

Future prospects

1. Further assignation of rays to actual chemical species.
2. By converting data into ratio or angular space, GMM model would not deal with absolute intensities.
3. Increase dimensionality of the problem:
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